

Q.24 Find the Moment of Inertia of a rectangle section 40 cm x 80 cm about (CO5)

- i) Horizontal axis passing through the centroid
- ii) Vertical axis through the centroid
- iii) Base

Q.25 Derive bending equation. (CO6)

No. of Printed Pages : 4
Roll No.

221821

2nd Sem / Mechanical (Tool & die Design)

Subject : Mechanics of Solids

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is a vector quantity? (CO1)

- a) Mass b) Volume
c) Density d) Force

Q.2 The rotational tendency of a force is called (CO2)

- a) Shear force b) Moment
c) Centroid d) Couple

Q.3 Which of the following methods applied for reducing friction? (CO3)

- a) Polishing b) Lubrication
c) Use of ball bearing d) All of the above

- Q.4 A line has centroid at its (CO4)
- a) Starting point b) End point
- c) Mid point d) None of the above
- Q.5 Which of the following materials is more elastic? (CO5)
- a) Rubber b) Glass
- c) Steel d) Wood
- Q.6 The moment of inertia of lamina is always minimum with respect to its (CO6)
- a) Base b) Centroidal axis
- c) Vertical axis d) Horizontal axis

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Volumetric strain. (CO1)
- Q.8 SI unit of moment is _____. (CO2)
- Q.9 Angle of friction is always _____ than 90° . (CO3)
- Q.10 Define Strain energy. (CO4)
- Q.11 Hooke's law holds good up to _____ limit. (CO5)
- Q.12 Define velocity ratio of a machine. (CO6)

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SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write the characteristics of a force. (CO1)
- Q.14 State the effect of a force on a body. (CO1)
- Q.15 State condition of equilibrium for bodies under coplanar non-concurrent force system. (CO2)
- Q.16 State the characteristics of couple. (CO2)
- Q.17 Write importance of friction. (CO3)
- Q.18 Explain the methods of reducing the friction. (CO3)
- Q.19 Differentiate between load and stress. (CO4)
- Q.20 State and explain theorem of perpendicular axis. (CO5)
- Q.21 State the assumption in the theory of simple bending. (CO6)
- Q.22 State law of machine. (CO7)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Draw stress-strain curve for a ductile material. Define factor of safety. (CO4)

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